

DISHA TAKAWALE

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PROFESSIONAL SUMMARY

Computer Engineering student with a 9.84/10 CGPA specializing in full-stack development, applied machine learning, and AI-powered applications. Experienced in building production-oriented MERN systems, Python ML services, REST APIs, OCR pipelines, and LLM-based agentic systems.

EDUCATION

Thadomal Shahani Engineering College

B.E. in Computer Engineering | CGPA: 9.84/10

Mumbai, MH

Aug 2024 – May 2027

Dr. Babasaheb Ambedkar Technological University

Diploma in Information Technology | CGPA: 9.34/10

Lonere, MH

Aug 2021 – Jun 2024

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

Frameworks/Libraries: React.js, Next.js, Node.js, Express.js, Flask, FastAPI, Tailwind CSS, Bootstrap, Scikit-Learn, OpenCV, RapidOCR, PyMuPDF

Databases: MongoDB, PostgreSQL

AI/LLM: Google Gemini API, LLM Function Calling, Multi-Agent Orchestration, Prompt Engineering

Tools: Git, GitHub, VS Code, Postman, MongoDB Compass, Figma, Google Antigravity

PROJECTS

Artha – Personal Finance Management Platform | *MERN, Python, Flask, Scikit-Learn*

Live Demo | GitHub

- Architected a microservices system separating a Node.js/Express CRUD backend from a Python ML service, orchestrating receipt uploads and dynamic category mapping via internal APIs.
- Built a native PDF parser (PyMuPDF) with a custom state-machine algorithm to extract itemized data from multi-page digital invoices, avoiding OCR overhead and fitting within a 512MB cloud tier; added OpenCV + RapidOCR fallback for scanned receipts.
- Built a dynamic category-matching engine (TF-IDF + Cosine Similarity) with synonym mapping to match items against arbitrary user-defined categories without retraining the classifier.

DetectiKid – ADHD Risk Screening Platform | *Next.js, TypeScript, FastAPI, Scikit-Learn*

GitHub

- Designed and built the frontend (Next.js 15, TypeScript, Tailwind CSS) for an interactive cognitive assessment game combining N-Back working memory and Go/No-Go impulse control tasks, in a four-person team.
- Built a real-time metric capture pipeline tracking reaction time, response accuracy, and omission errors during gameplay, feeding a FastAPI backend for ML inference.
- Implemented PDF report generation summarizing assessment metrics and classification results.

Engine Diagnostic AI Agent | *Python, Google Gemini, Multi-Agent Systems*

GitHub

- Built a 3-stage multi-agent pipeline (data ingestion, diagnostic reasoning, report generation) to analyze OBD-II vehicle telemetry and generate structured automotive fault reports.
- Implemented Gemini function calling with custom tool declarations to run diagnostic checks (thermal risk, RPM stability, sensor correlation) triggered dynamically by the LLM.
- Designed severity-based report synthesis (Critical/High/Medium/Low) with root-cause analysis and repair recommendations from structured agent outputs.

EXPERIENCE

UI/UX Designer

Feb 2025 – May 2025

Xonther

Remote

- Designed high-fidelity mobile interfaces and collaborated closely with developers to ensure accurate implementation within React Native constraints.
- Improved navigation flows and reduced interaction friction through iterative design improvements.

Internship Trainee

Jun 2023 – Jul 2023

Kopran Ltd.

Mahad, MH

- Updated and maintained internal product database systems to improve workflow efficiency.

CERTIFICATIONS

Full Stack Web Development Certification – Udemy

LEADERSHIP, ACTIVITIES & INTERESTS

NSS Joint Student Leader (2025-2026): Coordinated student volunteer initiatives and community service events at TSEC.

Hostel Secretary (2023–2024): Managed administrative operations and student grievances for a 100+ student hostel facility.

Technical Interests: Full Stack Development, UI/UX Design